

PROJECT REPORT

Healthcare Management System (HMS)

A PROJECT REPORT ON "MED-FLOW" HEALTHCARE SYSTEM

(Developed using Agile Scrum Methodology)

Submitted By: Neeraj Sahu

1. Abstract

The modern healthcare industry faces significant challenges in managing patient data, appointment scheduling, and doctor availability. Manual systems lead to errors, lost records, and long waiting times. This project, "**Med-Flow**," is a comprehensive Healthcare Management System designed to digitize these operations, ensuring better patient care and operational efficiency.

The project was developed using the **Agile Scrum Methodology**. Unlike traditional methods, Agile allowed us to prioritize critical modules like "Emergency Registration" and "Doctor Availability" early in the development cycle. The iterative nature of Agile ensured that the system remained adaptable to changing medical regulations and user feedback, ultimately delivering a secure, HIPAA-compliant platform.

Keywords: Agile, Healthcare, EMR (Electronic Medical Records), Patient Scheduling, Data Privacy.

2. Healthcare System Overview

2.1 Introduction

"Med-Flow" is a web-based application that connects Patients, Doctors, and Hospital Administrators. It serves as a central repository for **Electronic Medical Records (EMR)**, allowing doctors to view patient history instantly. It also streamlines the appointment booking process, reducing the burden on reception staff.

2.2 Problem Statement

Legacy healthcare systems often operate in silos (Reception has one system, Pharmacy has another).

1. **Data Fragmentation:** A doctor often doesn't know what medication another specialist prescribed.
2. **Scheduling Conflicts:** Manual booking leads to double-booking of slots.
3. **Security Risks:** Physical files are easily lost or accessed by unauthorized personnel.

2.3 The Agile Solution

Healthcare requirements are complex and sensitive. By using **Agile**, we broke the project into small, testable increments. We focused on building a secure "Patient Portal" first, ensuring that data privacy protocols were established before adding complex features like "Billing."

3. Agile Process Model

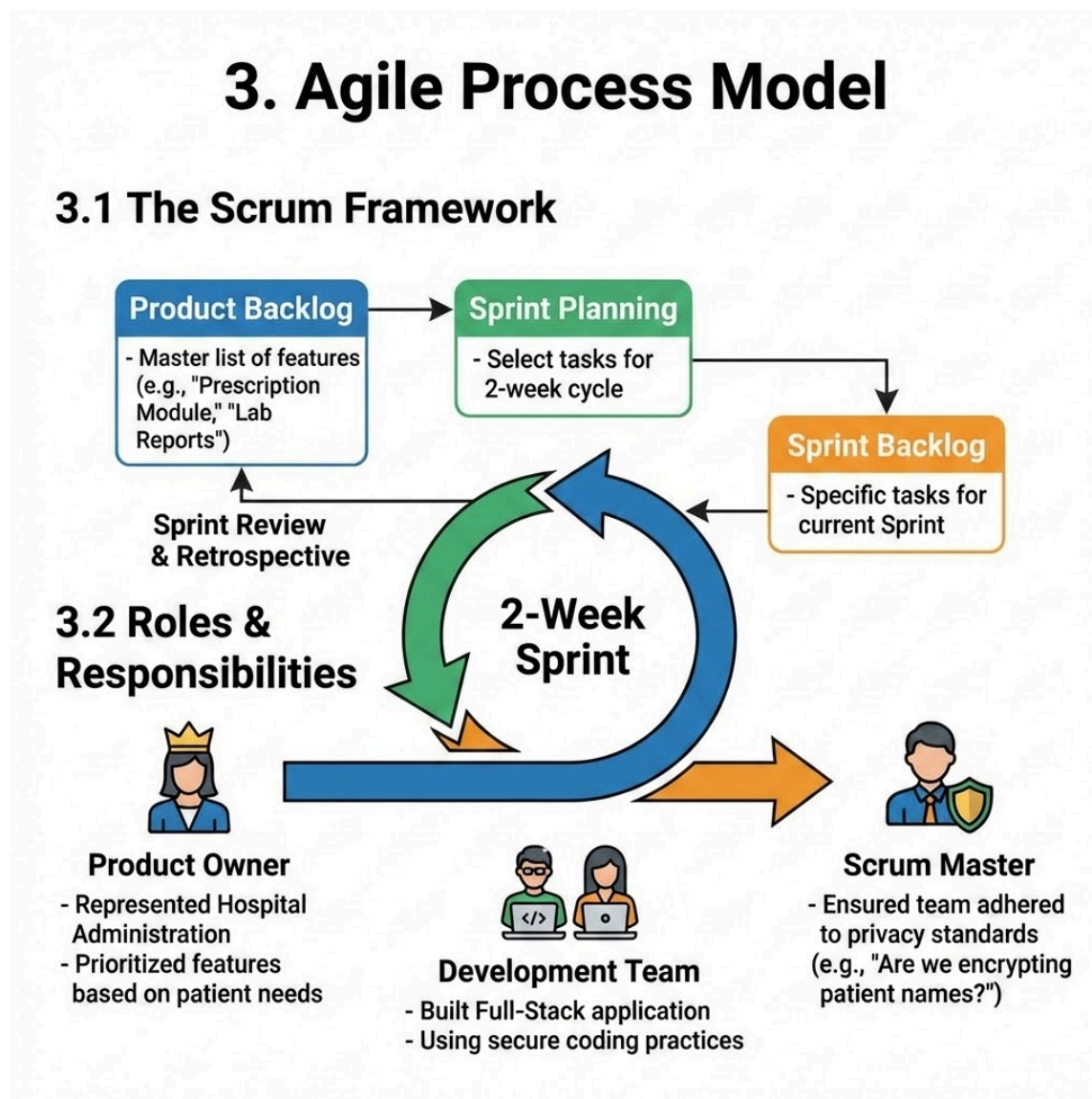
3.1 The Scrum Framework

We adopted the standard Scrum framework with 2-week Sprints.

- **Product Backlog:** A master list of features (e.g., "Prescription Module," "Lab Reports").
- **Sprint Backlog:** The specific tasks selected for the current 2-week cycle.

3.2 Roles & Responsibilities

- **Product Owner:** Represented the Hospital Administration. Prioritized features based on patient needs.
- **Scrum Master:** Ensured the team adhered to privacy standards (e.g., "Are we encrypting patient names?").
- **Development Team:** Built the Full-Stack application using secure coding practices.



4. User Stories (Requirements)

In Agile, requirements are captured as User Stories from the perspective of the end-user.

ID	Role	User Story	Acceptance Criteria
US-01	Patient	I want to book an appointment online.	System shows only available slots.
US-02	Doctor	I want to view a patient's medical history.	Access is granted only for appointed patients.
US-03	Admin	I want to add new doctors to the system.	Doctor ID and Specialization must be validated.
US-04	Receptionist	I want to generate a bill for the patient.	Bill must include consultation + medicine charges.

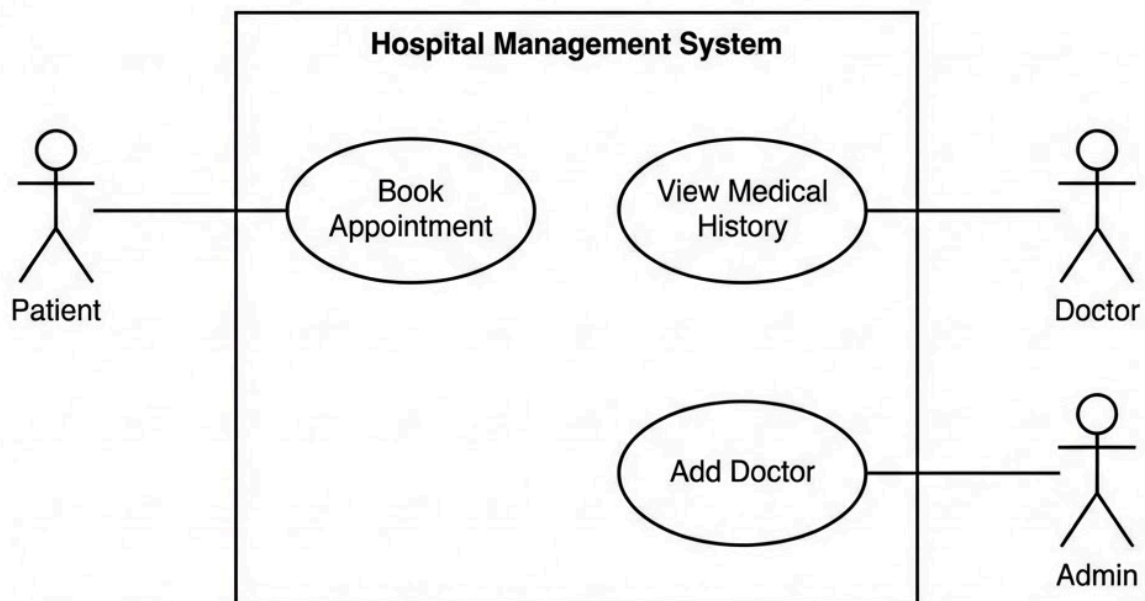


Figure 2: Use Case Diagram showing interactions between Patient, Doctor, and Admin.

5. Sprint-Based Development

Sprint 1: Registration & Security (Weeks 1-2)

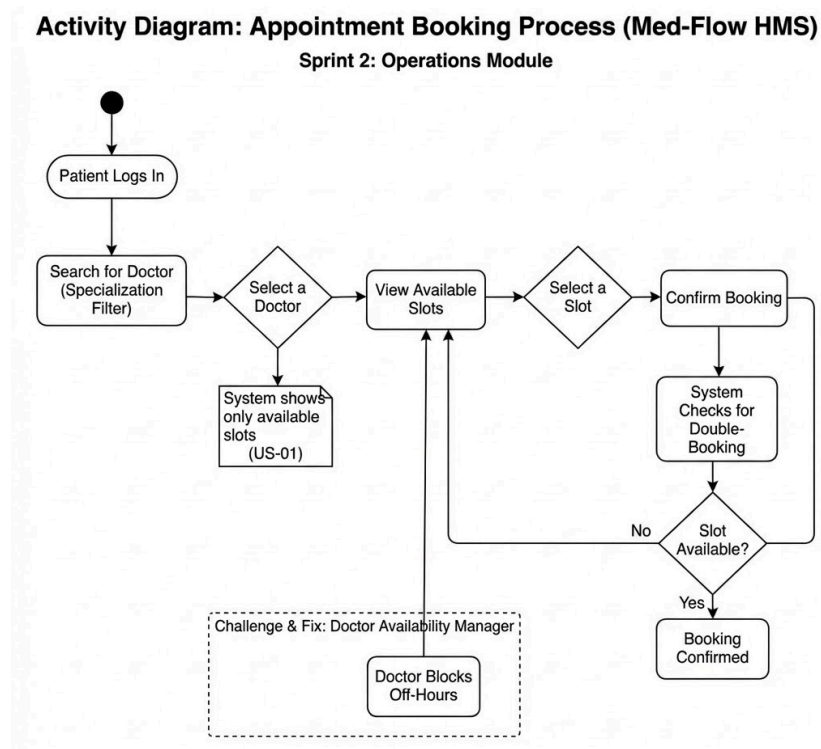
- **Focus:** Role-Based Access Control (RBAC).
- **Deliverables:**
 - Patient Registration with unique "Medical ID" generation.
 - Doctor Login (Doctors can see patients; Patients cannot see other patients).
- **Testing:** Validated that a Patient cannot access the Admin Dashboard (Vertical Privilege Escalation).

Sprint 2: Operations Module (Weeks 3-4)

- **Focus:** Appointment Scheduling.
- **Deliverables:**
 - "Find a Doctor" Search (Filter by Cardiology, Orthopedics, etc.).
 - Slot Booking Logic (Preventing double-booking).
- **Challenge:** Doctors have irregular schedules.
- **Fix:** Added a "Doctor Availability Manager" allowing doctors to block their off-hours.

Sprint 3: EMR & Diagnostics (Weeks 5-6)

- **Focus:** Medical Records.
- **Deliverables:**
 - Digital Prescription Pad (Doctor types meds, Patient downloads PDF).
 - Lab Report Upload.
- **Outcome:** A fully functional, paperless hospital workflow.



6. System Features

6.1 Electronic Medical Records (EMR)

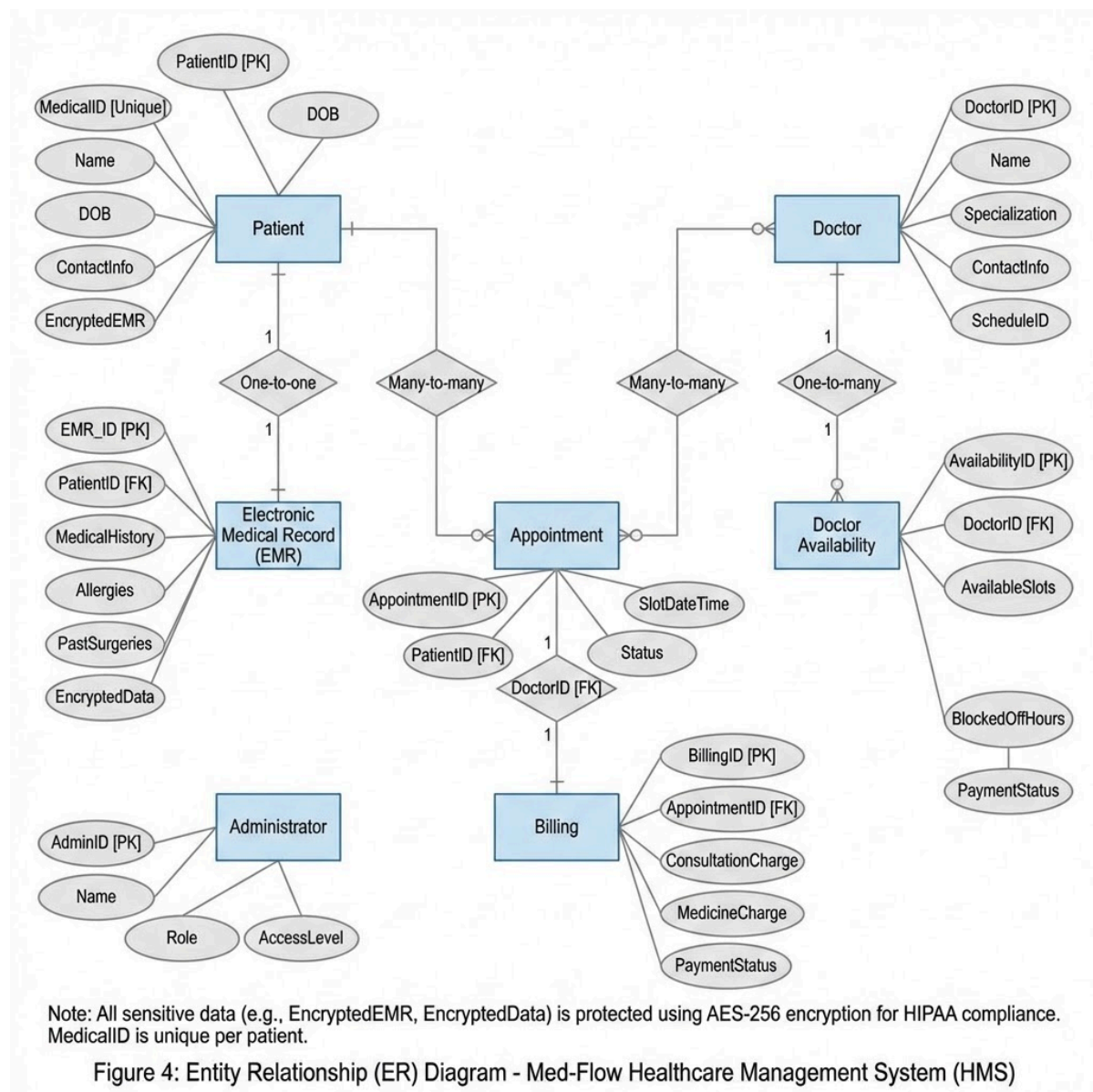
- Stores patient history, allergies, and past surgeries.
- **Privacy:** Data is encrypted at rest using AES-256 encryption.

6.2 Doctor Dashboard

- Shows a "Queue" of patients waiting for the day.
- Allows quick access to the "Prescribe Medicine" form.

6.3 Patient Portal

- Users can download prescriptions and bills.
- View upcoming appointment reminders.



7. Importance of Agile in Healthcare Software

7.1 Patient Safety First

In healthcare, a software bug can be life-threatening (e.g., wrong allergy info). Agile's **Continuous Testing** approach meant that every feature was tested daily. We didn't wait until the end to check if the "Allergy Alert" worked.

7.2 Regulatory Compliance

Healthcare laws (like HIPAA or local data protection acts) change. Agile allows the team to pivot. If a new law requires "Two-Factor Authentication" for doctors, we can add it in the next sprint without scrapping the whole project.

7.3 User Adoption

Doctors are busy. If the software is hard to use, they won't use it. By demonstrating the UI to doctors after Sprint 1, we received feedback to make the buttons larger and the text clearer, leading to higher adoption rates.

8. Conclusion

The "Med-Flow" project illustrates the effectiveness of **Agile Methodology** in critical sectors like healthcare. By breaking down the massive scope of hospital management into manageable sprints, we delivered a high-quality, secure, and user-friendly system. The project not only solved the problem of data fragmentation but also established a robust framework for future telemedicine integrations.